

Answer Key — Chemistry Grade 10

Sub-Strand 3.1: Acids and Bases • Lesson 4: Why is Lemon Juice Sour? Acids Release H^+ Ions in Water (Arrhenius Definition)

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
A	C	D	C	C	C	B	A

USE AFTER: OBSERVE PHASE — Part A (data): data card with pH and H^+ concentration

Q1. Pure water has pH 7.0 and H^+ concentration of 1×10^{-7} mol/L. Which statement about the H^+ ions in pure water is correct?

- A. It has a tiny amount of H^+ , far less than lemon juice**
- B. It has more H^+ than vinegar
- C. It has the same amount of H^+ as lemon juice
- D. It has no H^+ ions at all

Correct answer: A. 1×10^{-7} mol/L is very small but not zero, and it is much less than lemon juice (1×10^{-2} mol/L) or vinegar (about 1.6×10^{-3} mol/L).

USE AFTER: OBSERVE PHASE — Part A (data): 'How many times more H^+ does lemon juice have than vinegar?'

Q2. Lemon juice has pH 2.0 and $\text{H}^+ = 1 \times 10^{-2}$ mol/L. Vinegar has pH 2.8 and H^+ about 1.6×10^{-3} mol/L. About how many times more H^+ does lemon juice have than vinegar?

- A. About 0.8 times
- B. About 10 times
- C. About 6 times**
- D. About 60 times

Correct answer: C. Divide the concentrations: $(1 \times 10^{-2}) \div (1.6 \times 10^{-3}) = 6.25$, which is about 6 times. The pH difference of 0.8 does not mean 0.8 times more H^+ , because each pH unit is a 10-fold change.

USE AFTER: OBSERVE PHASE — Part B (simulation or paper model): acid molecules dissolving in water and releasing H^+ (aq) ions

Q3. In a particle simulation, acid molecules from lemon juice dissolve in water. What do learners see happen?

- A. The acid molecules stay whole and release nothing
- B. The acid molecules turn into sugar particles
- C. The acid molecules remove all the water particles
- D. The acid molecules release H^+ (aq) ions into the water**

Correct answer: D. An acid dissolving in water releases H^+ (aq) ions. This is the particle-level change that explains the properties of acids.

USE AFTER: EXPLAIN PHASE — Pairs' CER explanation and the class definition recorded on the board

Q4. Which statement is the Arrhenius definition of an acid?

- A. A substance that is harmful if swallowed
- B. A substance that contains no water
- C. A substance that produces H^+ (aq) ions when dissolved in water**
- D. A substance that tastes sour in the laboratory

Correct answer: C. The Arrhenius definition says an acid produces H^+ (aq) ions in water. Being harmful or tasting sour is not the definition, and we never taste chemicals in the laboratory.

USE AFTER: EXPLAIN PHASE — Claim-evidence-reasoning: 'Why is lemon juice sour, and why does it have a lower pH than vinegar?'

Q5. Lemon juice (pH 2.0) and vinegar (pH 2.8) are both acids. Why does lemon juice have the lower pH?

- A. It contains no water, but vinegar does
- B. It has fewer H^+ ions than vinegar
- C. It has a higher concentration of H^+ ions than vinegar**
- D. It contains more sugar than vinegar

Correct answer: C. A lower pH means a higher H^+ concentration. Lemon juice has about 6 times more H^+ ($1 \times 10^{-2} \text{ mol/L}$) than vinegar (about $1.6 \times 10^{-3} \text{ mol/L}$).

USE AFTER: DQB PHASE — Driving Question Board: linking notes to the hyperacidity question

Q6. Stomach acid is hydrochloric acid. Which particle does it release when it dissolves in water, making it an acid?

- A. No particles, because it is a liquid
- B. Only water particles
- C. H^+ (aq) ions**
- D. Sugar particles

Correct answer: C. Hydrochloric acid is an acid, so by the Arrhenius definition it releases H^+ (aq) ions in water, just like the acids in lemon juice and vinegar.

USE AFTER: MODEL BUILDING PHASE — Model page: three boxes for lemon juice, vinegar and water showing H^+ (aq) ions

Q7. In a Model drawing, lemon juice ($\text{H}^+ = 1 \times 10^{-2} \text{ mol/L}$) is shown with 6 H^+ dots. Which drawing for vinegar (about $1.6 \times 10^{-3} \text{ mol/L}$) and water ($1 \times 10^{-7} \text{ mol/L}$) is best?

- A. Vinegar 6 dots, water 0 dots
- B. Vinegar 1 dot, water almost no dots**
- C. Vinegar 1 dot, water exactly 0 dots
- D. Vinegar 12 dots, water 6 dots

Correct answer: B. Lemon juice has about 6 times more H^+ than vinegar, so 6 dots against 1 dot. Water has a tiny amount of H^+ , so it is drawn with almost none, not exactly zero.

USE AFTER: END OF LESSON — End of lesson: linking H^+ ions to the litmus and indicator colours from Lessons 2-3

Q8. Lemon juice turns blue litmus paper red. Which explanation uses the idea from this lesson?

- A. Lemon juice contains H^+ (aq) ions released by its acid in water**
- B. Lemon juice is red in colour
- C. Lemon juice has no water particles
- D. Lemon juice contains sugar particles

Correct answer: A. Acids release H^+ (aq) ions in water. These ions explain the low pH, the sour taste and the colour change of indicators such as litmus.